

## Powerful combination of long chain Ceramides and bioactive Ceramides – our premium solution for healthy aging

### Intended use

Active for skin care

### Benefits at a glance

- Powerful combination of 7 Ceramides in a modern emulsifier system without listed preservatives.
- Combining skin identical long chain Ceramides with short chain bioactive Ceramides unfolds a product perfectly suitable for aging skin.
- Improved barrier function from within and regeneration by activation of the skin's natural water management system
- Modern, alternative preservation system
- China IECIC listed ingredients

### INCI (PCPC name)

Ceramide EOP; Ceramide NP; Ceramide NS; Ceramide AP; Cholesterol; Behenic Acid; Polyglyceryl-10 Stearate; Polyglyceryl-6 Behenate; Glycerin; Cetearyl Alcohol; Glyceryl Stearate; Sodium Cetearyl Sulfate; Triethylcitrate; Lactic Acid, Aqua/Water

### Chemical and physical properties (not part of specifications)

|               |                    |
|---------------|--------------------|
| Form          | Low viscous liquid |
| Active matter | approx. 2%         |

### Properties

Due to demographic changes the amount of people aged 50 years or older will increase significantly in the upcoming years. This leads to a strong demand for skin care products addressing the specific needs of this population.

Aging skin is characterized by a reduced cell metabolism which results in a defective skin surface leading to moisture loss and dry skin accompanied by a loss of elasticity and increase of wrinkled appearance.

With our SKINMIMICS® technology we offer multi-lamellar dual action lipid blends based on skin identical Ceramides. It combines skin identical long chain Ceramides with bioactive short-chain Ceramides (SPHINGOKINES®), vegetable cholesterol and behenic acid.

The Ceramides are based on Evonik's Ceramide expertise: the necessary Ceramide building blocks (phytosphingosine and sphingosine) are produced by a fermentative process, which yields the skin identical Ceramide stereochemistry.

Recognizing the specific needs of aging and especially mature skin, the SKINMIMICS® product range is specially designed to support the healthy aging of skin.

SKINMIMICS® PRO MB is a next generation of our SKINMIMICS® technology, embedding the Ceramides in a modern, natural derived, multi-lamellar formulation chassis. In contrast to SKINMIMICS® MB SKINMIMICS® PRO MB does not contain any listed preservative and ethoxylates and all ingredients are China registered.

|                  | SKINMIMICS® PRO MB                                                                                                                                    | SKINMIMICS® MB                                                                                                                                                        |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Emulsifier       | Polyglycerin-10 Stearate,<br>Polyglycerin-6 Behenate                                                                                                  | Ceteareth-25                                                                                                                                                          |
| Stabilizer/Waxes | Cetearyl Alcohol<br>Glyceryl Stearate<br>Glycerin<br>Sodium Cetearyl Sulfate                                                                          | Cetyl Alcohol<br>Glyceryl Stearate<br>Glyceryl Oleate<br>Lecithin<br>Glycerin<br>Tocopherol<br>Ascorbyl Palmitate                                                     |
| Preservative     | Triethylcitrate<br>Lactic Acid                                                                                                                        | Phenoxyethanol<br>Ethylhexylglycerin                                                                                                                                  |
| Skin lipids      | Ceramide EOP<br>Ceramide NP 16<br>Ceramide NP 22<br>Ceramide NS<br>Caprooyl Phyto-sphingosine<br>Caprooyl Sphingo-sine<br>Cholesterol<br>Behenic Acid | Ceramide EOS<br>Ceramide EOP<br>Ceramide NP 16<br>Ceramide NP 22<br>Ceramide NS<br>Caprooyl Phyto-sphingosine<br>Caprooyl Sphingo-sine<br>Cholesterol<br>Behenic Acid |

Tab. 1: Ingredients of SKINMIMICS® PRO MB in comparison to SKINMIMICS® MB

The various Ceramides and skin lipids contained in SKINMIMICS® PRO MB have specific roles and functions in the skin barrier:

| Ceramide            | Function                                                                                                                                                       |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ceramide EOP        | The anchor bolt.<br>It forms and stabilizes the lipid formation of the lipid lamellae and connects to the cornified envelope. Connector between lipid bilayers |
| Ceramide NP         | The care-taker<br>Abundant in our lipid lamellae, it protects against moisture loss                                                                            |
| Ceramide NS         | The flexible care-taker<br>It brings flexibility into the lipid lamellae                                                                                       |
| Ceramide AP         | The normalizer<br>It plays a key role in the desquamation process of dead keratinocytes                                                                        |
| Bioactive Ceramides | The signal transmitters<br>They function in biochemical regulation of cell growth, differentiation and pathways in skin aging processes.                       |
| Free fatty acids    | The essential fillers<br>They impact the skin barrier integrity of the intercellular lipids                                                                    |
| Cholesterol         | The organizer<br>It is essential for the correct skin barrier function, contributes to the right lamellar and lateral organization                             |

Tab. 2: Function of different skin lipids of SKINMIMICS® PRO MB

## Efficacy studies

### *Ex vivo* study on human skin explants

For this study skin explants of a 53 years old, caucasian women were used. The skin explants were treated for 6 days with the test formulations. As test formulation a basic O/W emulsion was used containing no active ingredient (vehicle), 5% SKINMIMICS® PRO MB or for comparison reasons 5% SKINMIMICS® MB. After 6 days treatment an immunostaining of Ceramides was performed using a monoclonal anti-Ceramides antibody (Glycobiotech, ref. MAB\_0013). For semi-quantification of the Ceramides in the stratum corneum ImageJ was used. With this image analysis software, the Ceramide content was determined by calculating the relative intensity of the color signal of the stained Ceramides.

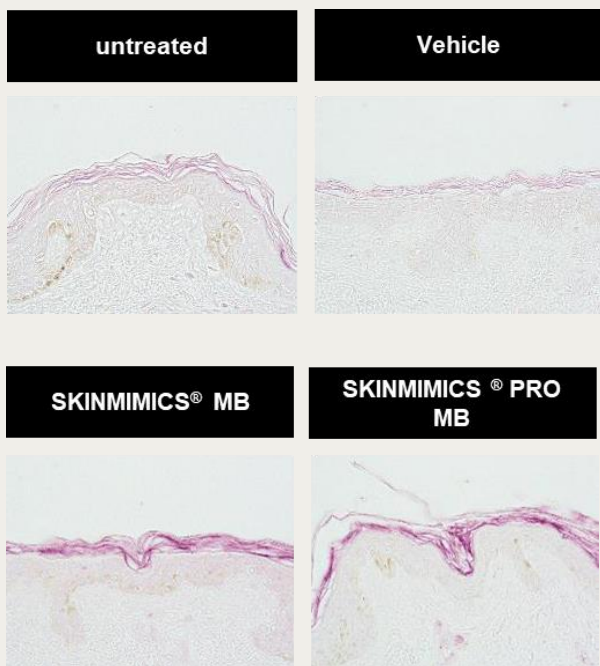


Fig. 1 Immunostaining of Ceramide in the stratum corneum

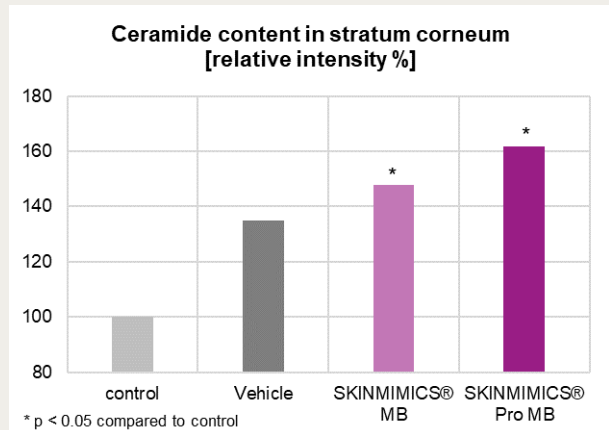


Fig. 2: Ceramide content in the stratum corneum

The images of the skin explants (Fig. 1) as well as the graph (Fig. 2) show clearly that SKINMIMICS® PRO MB showed a visible increase of Ceramide concentration in the Stratum Corneum. The efficacy of SKINMIMICS® PRO MB and SKINMIMICS® MB is comparable.

### ***In vivo* skin biopsy study**

10 panellists aged between 35–63 years applied a simple serum formulation with 5% SKINMIMICS® MB for 4 weeks on the buttock. A part of the buttock was not treated and used as control. After 4 weeks skin biopsies were taken from the treated as well as the untreated skin and analyzed by qRT-PCR for different skin markers. The panelists were divided into two age groups, younger or older than 50 years. The results were calculated for these two age groups.

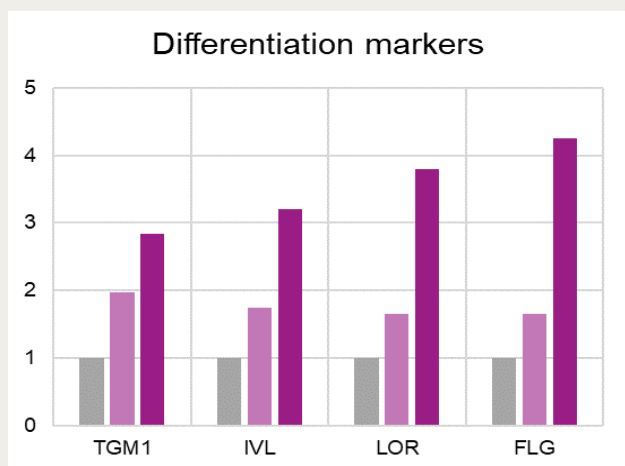


Fig. 3: Upregulation of differentiation markers

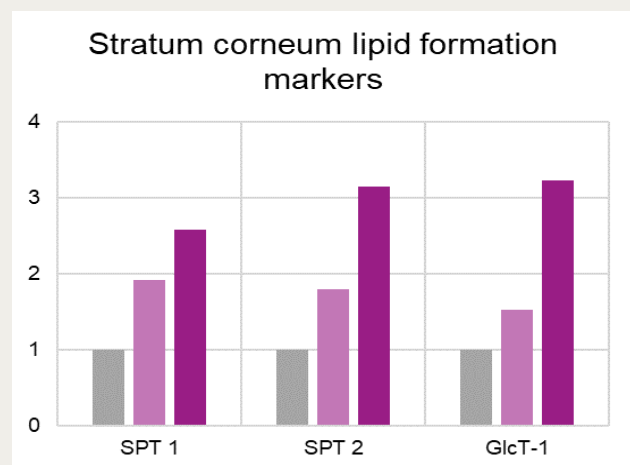


Fig. 4: Upregulation of Stratum corneum lipid formation markers

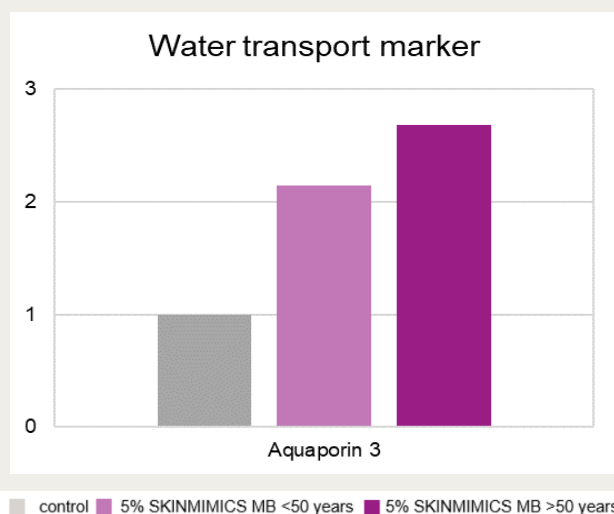


Fig. 5: Upregulation of Aquaporin 3

The upregulation of the skin markers shown in Fig. 3–5 showed that SKINMIMICS® MB improves barrier function, supplies the skin with moisture from within, activates the skin's natural water management and induces lipid formation.

Regarding the two age groups it is obvious that the markers are stronger upregulated in the older group. This makes SKINMIMICS® MB especially suitable for people above 50 years by supporting the healthy and well-aging of the skin.

Due to the nearly identical Ceramide composition of SKINMIMICS® and SKINMIMICS® PRO MB similar results are expected with SKINMIMICS® PRO MB.

## Formulation hints

SKINMIMICS® PRO MB is dispersible in water and can be cold processed.

Preparation of O/W emulsions: SKINMIMICS® PRO MB should be added during the cooling process at temperatures below 40°C. If necessary, it is also possible to add SKINMIMICS® PRO MB directly to the water phase prior to the homogenization step.

It is possible to replace SKINMIMICS® MB by SKINMIMICS® PRO MB in O/W emulsions without adaptation of the formulation or at least with very minor adaptations.

Preparation of W/O emulsions: SKINMIMICS® PRO should be added during the cooling process prior to the second homogenization step. Due to the O/W emulsifier in SKINMIMICS® PRO MB phase inversion or instabilities of the W/O emulsion can occur. This can be prevented by using a sufficient amount of suitable W/O emulsifiers like ABIL® EM 90, ISOLAN® GPS or ISOLAN® PDI

## Application

SKINMIMICS® PRO MB is suitable for all kind of skin care formulations addressing the needs of aging and mature skin:

- Moisturizing products
- Face care products
- Anti-aging and anti-wrinkle products
- Skin repair
- Skin protection

## Recommended usage concentration of SKINMIMICS® PRO MB

0.5–5% (clinical studies ongoing)

## Guideline Formulations

If you are interested in guideline formulations, please visit our homepage <https://personal-care.evonik.com>.

## Hazardous goods classification

Information concerning

- classification and labelling according to regulations for transport and for dangerous substances
- protective measures for storage and handling
- measures in case of accidents and fires
- toxicity and ecological effects

is given in our material safety data sheets.

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